

Study Shows How Earth's Orbit Affects Ice Ages

A recent study suggests the next ice age will likely begin in about 11,000 years unless emissions change the effects of natural climate cycles.

An ice age is a period of reduced temperatures causing ice to cover large areas of the northern and southern parts of the world.

A group of scientists studied changes in the shape of the Earth's orbit around the sun, the tilt of the Earth's axis and the movement of the axis itself, called a wobble. These three motions are believed to happen in cycles. The Serbian scientist Milutin Milankovitch developed a theory about 100 years ago that these cycles affected Earth's long-term climate.

The team noted that small changes in Earth's orbit around the Sun caused predictable cycles of warming and cooling over a period of about 100,000 years.

These findings permitted the team to determine how the three factors of tilt, wobble, and the shape of Earth's orbit might have affected the Earth's climate over the last 900,000 years.

The group published its findings in Science on February 27.

The scientists examined a million-year record of climate by looking at ice sheets across the northern half of the world and deep ocean temperatures.

They then compared this data with small but regular changes in Earth's orbit over time.

The lead writer of the study is Stephen Barker, a professor at Cardiff University in Britain. He told the French News Agency (AFP) that for many years it has been difficult for scientists to show exactly how small changes in Earth's orbit have led to huge changes between warm and cold periods.

Earth is believed to go through cycles of ice ages and warm periods. The last ice age is thought to have ended around 11,700 years ago. Scientists, such as Milankovitch, have long seen a connection between Earth's orbit and climate.

FILE - A glacier in Half Moon Bay, Antarctica, February 18, 2018. (Reuters/Alexandre Meneghini)

However, researchers have struggled to precisely date when each of the climate changes happened. It has also been difficult for them to know exactly which orbit positions affect the start and stop of cold or warm periods.

Barker said his team of researchers studied slow changes in temperature over long periods of time instead of only looking at shorter periods of change between cold and warm climates.

Barker told AFP, "We would expect a glaciation to occur within the next 11,000 years, and it would end in 66,000 years' time."

Lorraine Lisiecki is a professor at the University of California, Santa Barbara, and a co-writer of the study. She said the study shows that changes in climate over tens of thousands of years are not random.

The team aims to expand its findings to investigate the long-term effects of human activity on climate. They also want to find out how it might affect Earth's natural climate cycles.

I'm Andrew Smith.

Issam Ahmed wrote this story for Agence-France Presse. Andrew Smith adapted it for VOA Learning English.

Vocabulary

emissions - n. พหุพจน์ ก๊าซที่ปล่อยสู่ชั้นบรรยากาศ

tilt - n. การเอียง, การลาดเอียง (ไม่ได้ตั้งตรง แต่เอียงไปด้านข้างเป็นมุม)

axis - n. แกน (เส้นสมมุติที่โลกหมุนรอบตัวเองหรือหมุนรอบวัตถุอื่น)

cycle - n. วัฏจักร, รอบ (การกระทำหรือชุดของการกระทำที่เกิดขึ้นซ้ำในช่วงเวลาปกติ)

sheet - n. แผ่น (วัสดุที่แบนและกว้าง เช่น กระจกหรือน้ำแข็ง)

precisely - adv. อย่างแม่นยำ, อย่างแน่นอน

factor - n. ปัจจัย (บางสิ่งบางอย่างที่ส่งผลกระทบต่อกระบวนการหรือการพัฒนา)

wobble - n. การโคลงเคลง, การส่าย (การเคลื่อนไหวที่ไม่มั่นคงจากด้านหนึ่งไปอีกด้านหนึ่ง)

glaciation - n. ยุคน้ำแข็ง, การเกิดธารน้ำแข็ง (ช่วงเวลาที่น้ำแข็งขนาดใหญ่ก่อตัวและเคลื่อนที่)

random - adj. สุ่ม, ไม่มีแบบแผน (ไม่แสดงลำดับหรือช่วงเวลาที่น่าเชื่อถือ)

Reading Comprehension Quiz

1. What is the main prediction of the recent study mentioned at the beginning of the article?

- A) The next ice age has already begun.
- B) Human emissions will completely prevent the next ice age.
- C) The next ice age is expected to start in approximately 11,000 years.
- D) Natural climate cycles are no longer relevant to the onset of ice ages.

2. How is an 'ice age' fundamentally defined in the article?

- A) A time when the Earth's orbit changes dramatically.
- B) A period of significantly lower temperatures leading to widespread ice cover.
- C) A cycle of warming and cooling over exactly 100,000 years.
- D) A natural climate cycle that is solely affected by human emissions.

3. Which of the following motions was NOT explicitly mentioned as being studied by the scientists in relation to Earth's climate?

- A) The tilt of the Earth's axis.
- B) The Earth's rotation speed.
- C) The wobble of the Earth's axis.
- D) The shape of the Earth's orbit around the sun.

4. What significant theory did the Serbian scientist Milutin Milankovitch develop about 100 years ago?

- A) A theory explaining the precise dating of all past climate changes.
- B) A theory that Earth's natural cycles affect its long-term climate.
- C) A method for examining a million-year record of climate.
- D) A prediction about the exact end of the next glaciation period.

5. According to the team's findings, what is the approximate duration of the predictable cycles of warming and cooling caused by small changes in Earth's orbit?

- A) Approximately 11,000 years.
- B) About 66,000 years.
- C) Roughly 900,000 years.
- D) Around 100,000 years.

6. The study's findings allowed the team to understand the climate impact of which three factors over the last 900,000 years?

- A) Solar flares, volcanic activity, and deforestation.
- B) Human emissions, atmospheric pressure, and ocean currents.
- C) The tilt, wobble, and shape of Earth's orbit.
- D) Global warming, rising sea levels, and urban expansion.

7. In which publication did the group publish its findings on February 27?

- A) The French News Agency (AFP)
- B) Cardiff University's internal journal

- C) Science
- D) VOA Learning English

8. To examine a million-year record of climate, what primary sources of data did the scientists use?

- A) Satellite images and ancient human records.
- B) Tree rings and fossilized plant remains.
- C) Ice sheets across the northern hemisphere and deep ocean temperatures.
- D) Geological strata and volcanic rock analysis.

9. What did the scientists compare their gathered climate data with?

- A) Predictions from ancient prophecies.
- B) Current rates of human-induced climate change.
- C) Small but regular changes in Earth's orbit over time.
- D) Data from atmospheric CO₂ levels from other planets.

10. According to Stephen Barker, what challenge have scientists faced for many years regarding Earth's orbit and climate?

- A) Identifying the precise composition of ice sheets.
- B) Showing exactly how small orbital changes lead to huge climate shifts.
- C) Confirming the existence of past warm periods.
- D) Predicting the next volcanic eruption accurately.

11. When is the last ice age generally thought to have ended?

- A) Approximately 100 years ago.
- B) Around 11,700 years ago.
- C) In the next 11,000 years.
- D) Exactly 66,000 years ago.

12. What specific difficulties have researchers encountered concerning the history of climate changes?

- A) They struggled to obtain funding for their research.
- B) They found it hard to gather enough samples of ice sheets.
- C) They struggled to precisely date climate changes and identify which orbit positions affect their start/stop.
- D) They were unable to collaborate effectively with international teams.

13. What was the distinguishing characteristic of Barker's team's research methodology?

- A) They focused exclusively on short-term climate fluctuations.
- B) They analyzed rapid and sudden temperature shifts.
- C) They studied slow temperature changes over long periods.
- D) They disregarded temperature data, focusing only on atmospheric composition.

14. Based on Stephen Barker's prediction, when is the next glaciation expected to conclude?

- A) Within the next 11,000 years.
- B) Approximately 11,700 years from now.
- C) In 66,000 years' time.
- D) After 900,000 years.

15. According to Lorraine Lisiecki, what does the study reveal about climate changes over tens of thousands of years?

- A) They are entirely random and unpredictable.
- B) They are primarily caused by recent human industrial activity.
- C) They exhibit a non-random, patterned behavior.
- D) They have no discernible connection to Earth's natural cycles.